

Evidence-Based Study Strategies

A practical learning-science lecture about durable recall, useful effort, and honest evidence of mastery.

RECALL

SCHEDULE

CORRECT

Learning goal

Choose study actions that still work when the page is closed.

Retrieval beats passive review

Close the material. Pull the answer from memory. Then check.

WHY IT WORKS

Strengthens access to knowledge and exposes gaps that familiarity can hide.

WHAT COUNTS

Low-stakes quizzes, blank-page recall, flashcards, and explaining without slides.

WHAT DOES NOT

Viewing the answer immediately and recognizing it is not retrieval practice.

LEARN



RETRIEVE



CHECK



RESTDY THE GAP

Space practice, then schedule it

Spacing distributes practice across separate sessions. A practice schedule turns that principle into a repeatable plan.



SPACING EFFECT

Some forgetting makes the next successful retrieval effortful and useful.

ADAPT THE INTERVAL

After a failure, return sooner. After easy recall, lengthen the interval.

Difficulty is useful only when it teaches

DESIRABLE DIFFICULTY

Harder now, better retention or transfer later.

PRODUCTIVE FRICTION

- Generation before explanation
- Spaced retrieval and varied examples
- Achievable effort with correction

MERELY DIFFICULT

- Confusing or ambiguous instructions
- A task beyond current prerequisites
- Effort with no useful feedback

Interleaving works by contrast

BLOCKED

A A A B B B C C C

INTERLEAVED

A B C A C B B A C

THE MECHANISM: DISCRIMINATION

Notice features > **Identify the problem** > **Choose a strategy**

Interleaving can lower immediate accuracy while improving later strategy selection and transfer. Smooth blocked practice may be misleading.

Feedback timing closes the loop

Give feedback after a genuine attempt, but soon enough to correct the mental model before an error becomes stable.

1

ATTEMPT

2

FEEDBACK

3

CORRECTION

4

RETRY

USE SOONER

Novices, factual errors, and misconceptions that could persist.

A BRIEF DELAY CAN HELP

Complex judgments where reflection is useful - never leave errors uncorrected.

Fluency is not mastery

FEELS EASY

Recognition

The page looks familiar. Each pass is faster. Highlighted sentences feel known.

Rereading alone: low utility

BUILDS DURABLE ACCESS

Retrieval

The material is closed. You produce the idea, check it, and repair a specific gap.

Practice testing: stronger evidence

Use rereading for orientation or targeted restudy - not as proof that learning happened.

A weekly system for durable learning

MON

Learn + retrieve

Close the slides and explain the core idea.

TUE

Short return

Retrieve again; shorten intervals after misses.

THU

Interleave

Mix related choices and name why each fits.

SUN

Cumulative check

Quiz old and new material, then correct gaps.

THE CHECK

**Can I still
retrieve the idea
and choose when to
use it with the
page closed?**

EVIDENCE, NOT EASE

Retrieve. Space. Discriminate. Correct.